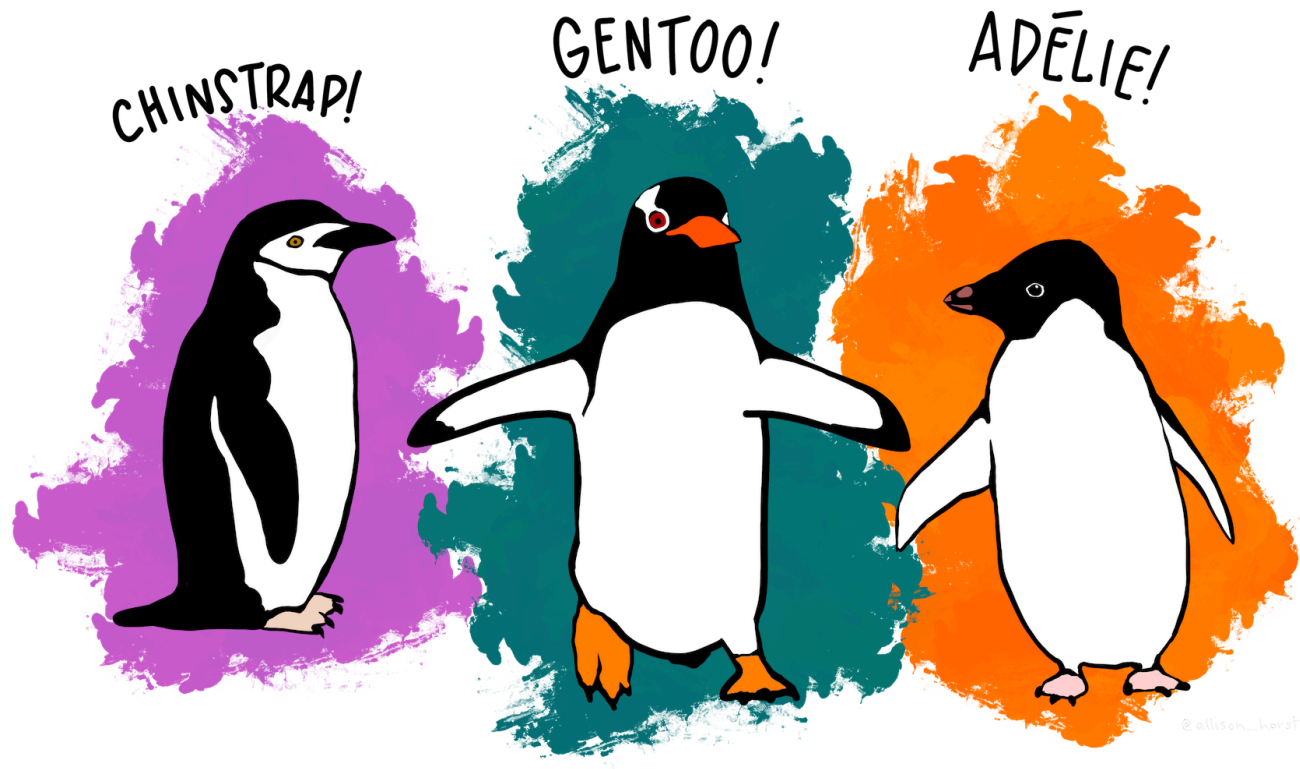


Data visualization

Agenda

- + Today: data visualization with `ggplot`
 - + Formalize what we've seen in class activities
- + HW 1 due Friday
 - + I'm available in office hours or by email if you have any questions
- + First part of project (choosing a dataset) released later this week
- + No class on Wednesday (I will be out in the morning)
 - + I will be back for office hours on Wednesday afternoon!

Data for today



Data visualization with ggplot2

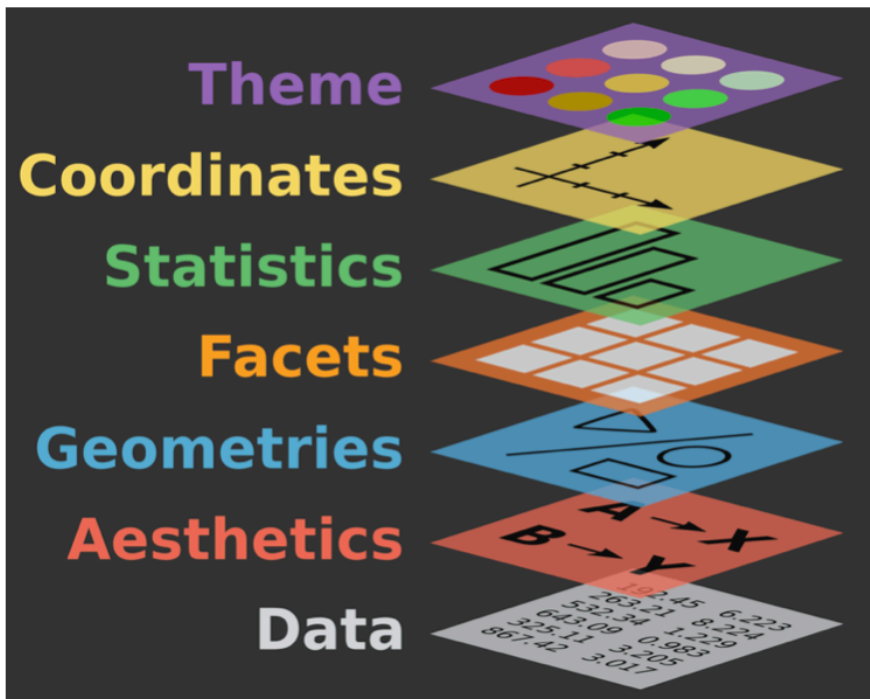
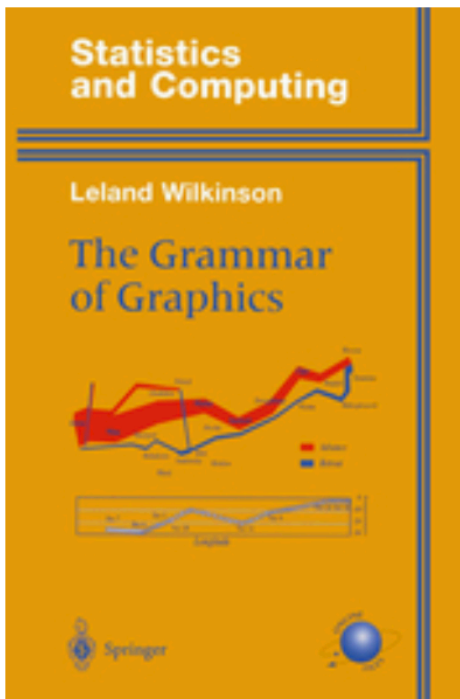
- + `ggplot2`: the R package
- + `ggplot`: the function (from `ggplot2`) used to make plots
- + `gg` stands for "Grammar of Graphics"



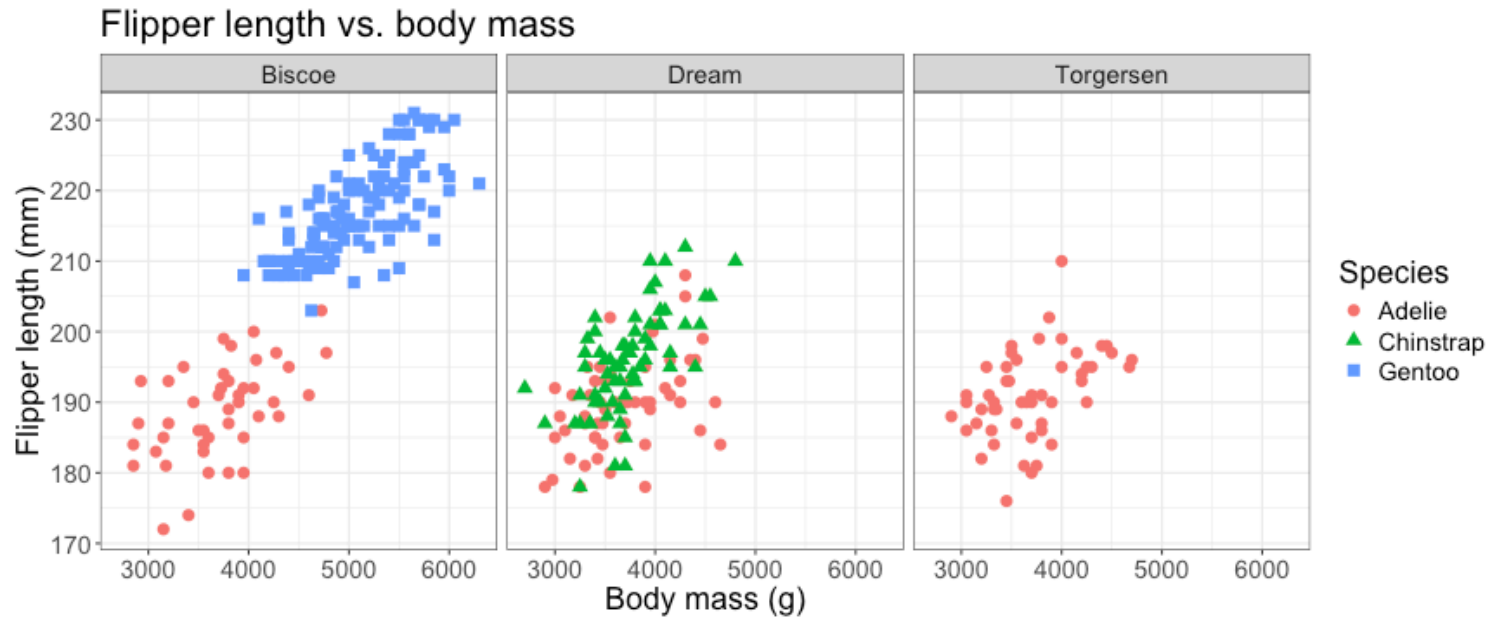
Artwork by @allison_horst

Grammar of Graphics

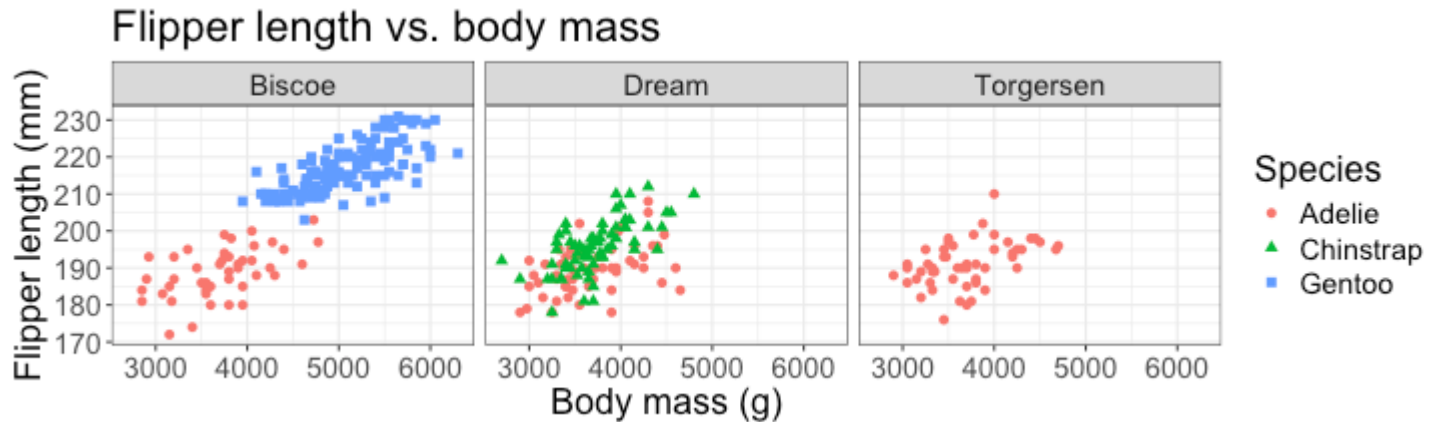
Build visualizations in layers:



Plot to make

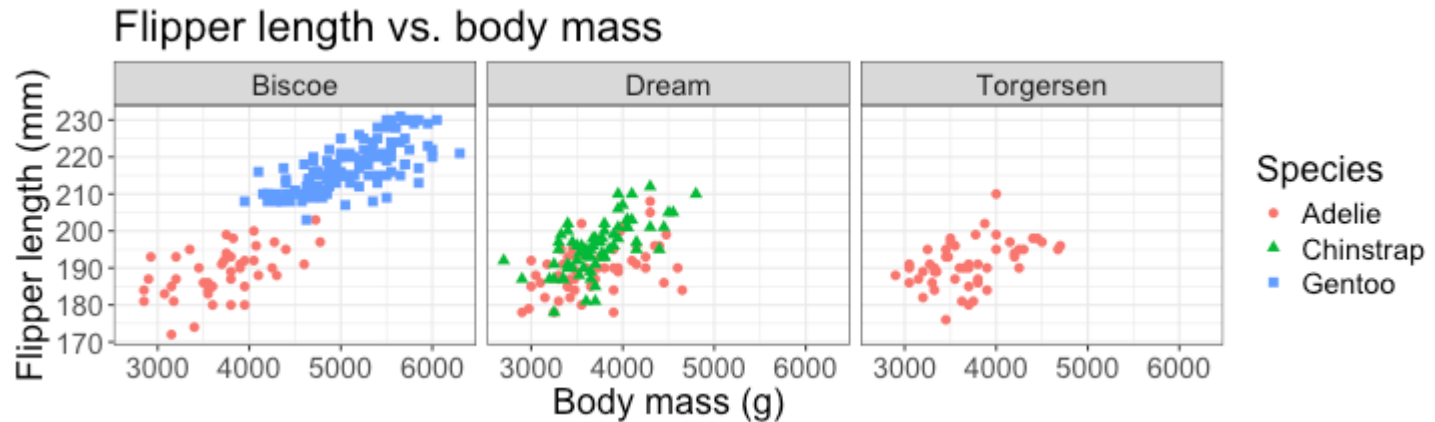


Layer 1: the data



Data: Which data do I want to plot?

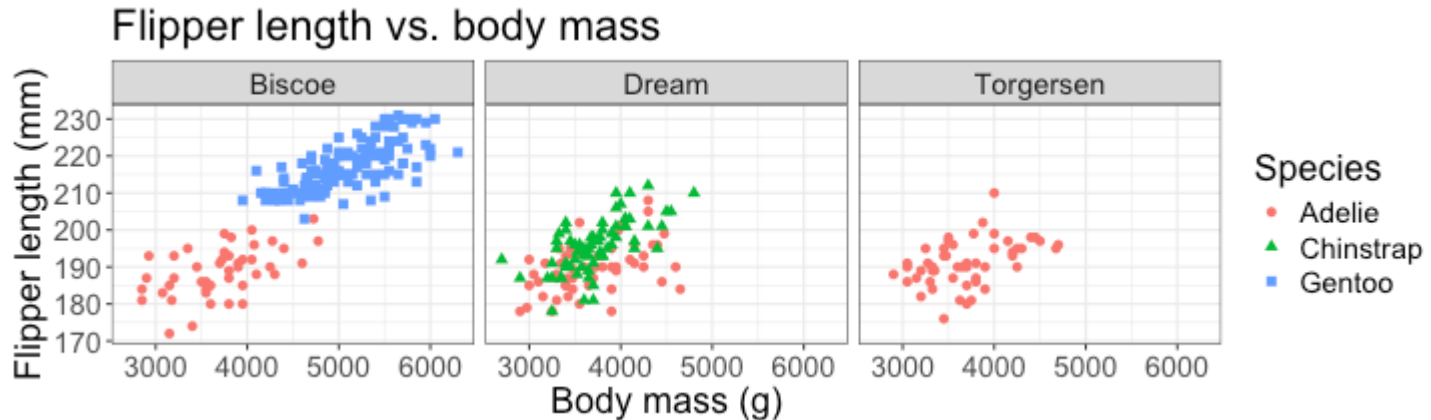
Layer 1: the data



Data: Which data do I want to plot?

penguins

Layer 2: Aesthetics

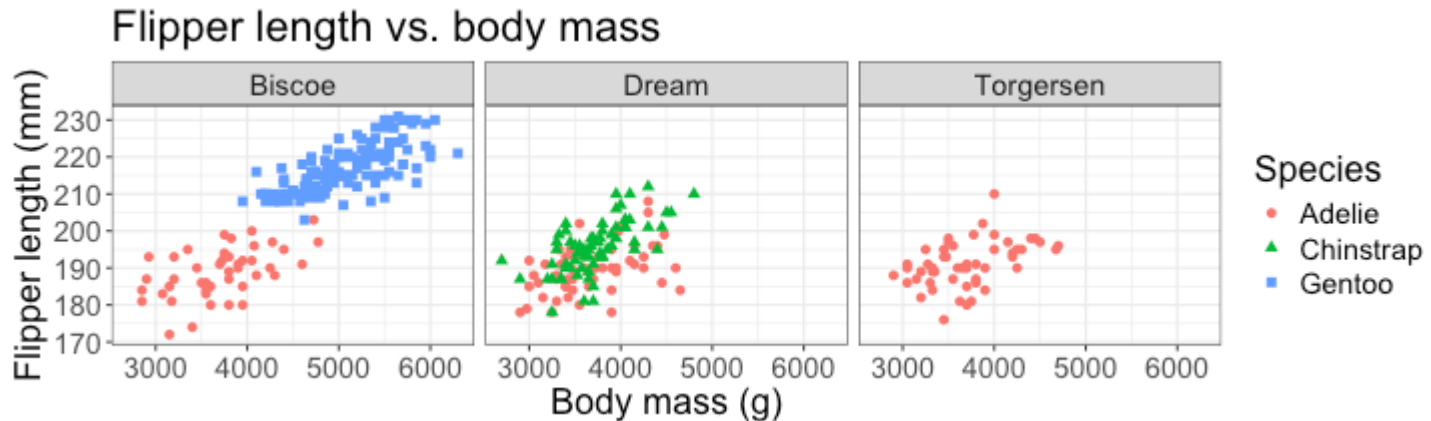


Aesthetics: mapping features of the plot to variables in the data

Which variables from the data do I want to plot?

- Body mass → x axis
- Flipper length → y axis
- Species → color, shape
- Island (one panel per island) (handle later)

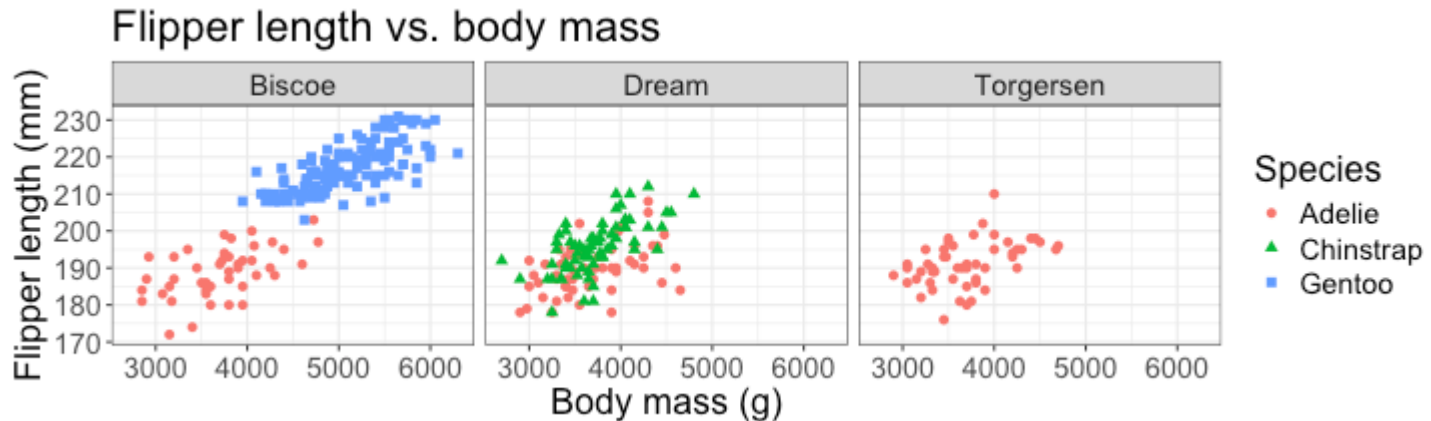
Layer 2: Aesthetics



Aesthetics: mapping features of the plot to variables in the data

```
penguins |> aesthetics  
  ggplot(aes(x = body_mass_g,  
             y = flipper_length_mm,  
             color = species,  
             shape = species))
```

Layer 3: Geometric objects

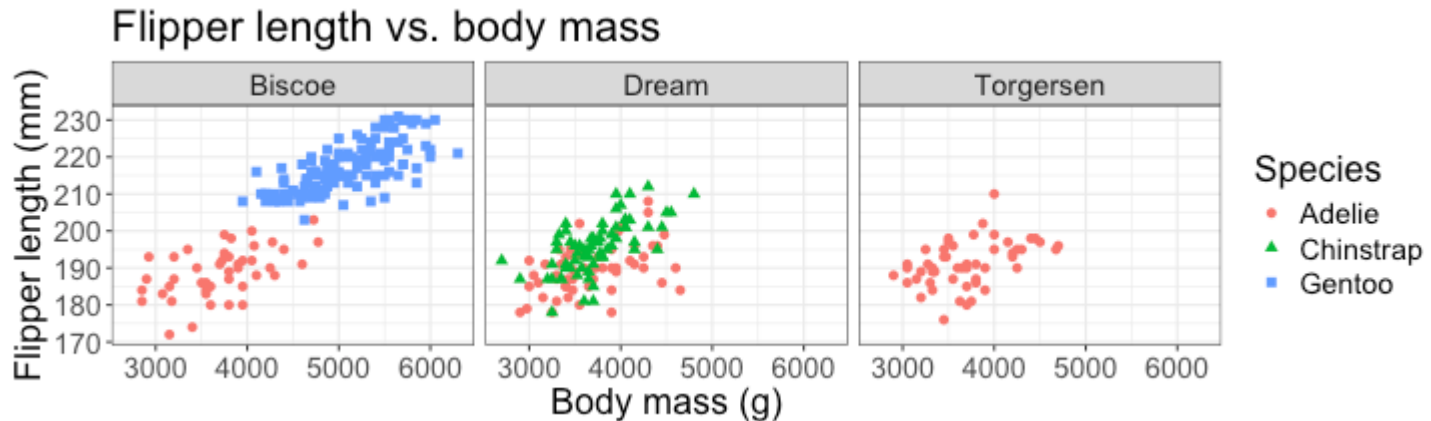


Geometric objects: objects we use to visualize the data

What objects do we use to represent the penguins on this plot?

By points — each point corresponds to a penguin

Layer 3: Geometric objects

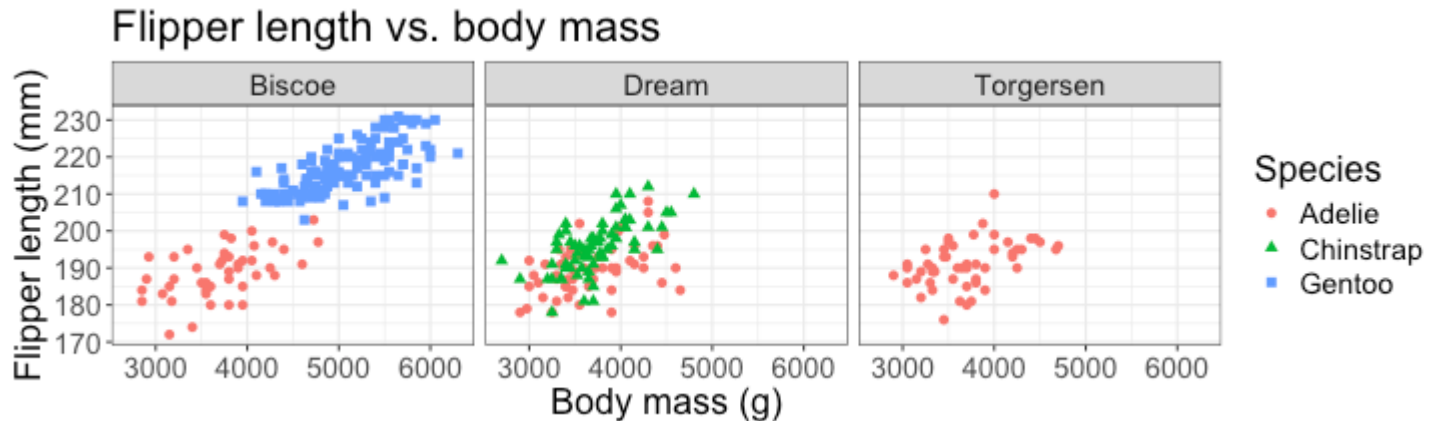


Geometric objects: objects we use to visualize the data

```
penguins |>  
  ggplot(aes(x = body_mass_g,  
             y = flipper_length_mm,  
             color = species,  
             shape = species)) (+) ← add layers together  
  geom_point()
```

↑
geometric object

Layer 3: Geometric objects



```
penguins |>  
  ggplot(aes(x = body_mass_g,  
             y = flipper_length_mm,  
             color = species,  
             shape = species)) +  
  geom_point()
```

Other geometric objects include `geom_histogram`, `geom_boxplot`, `geom_bar`, `geom_smooth`, and `geom_line`

What we have so far

```
penguins |>  
  ggplot(aes(x = body_mass_g,  
             y = flipper_length_mm,  
             color = species,  
             shape = species)) +  
  geom_point()
```



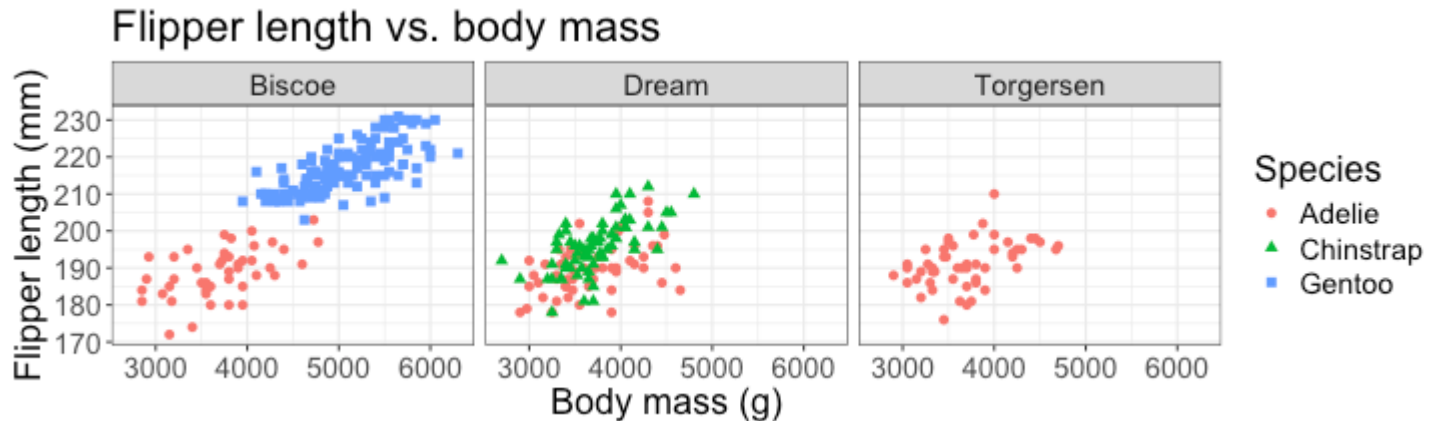
What we have so far



Still need to:

- + Divide the plot by island (Biscoe, Dream, and Torgersen)
- + Add labels and a title
- + Change the background

Layer 4: Faceting



Facets: split visualization by the value of another variable

```
penguins |>
  ggplot(aes(x = body_mass_g,
             y = flipper_length_mm,
             color = species,
             shape = species)) (+)
  geom_point() (+)
  facet_wrap(~island)           ← facet by island
```

theme_classic

theme_bw

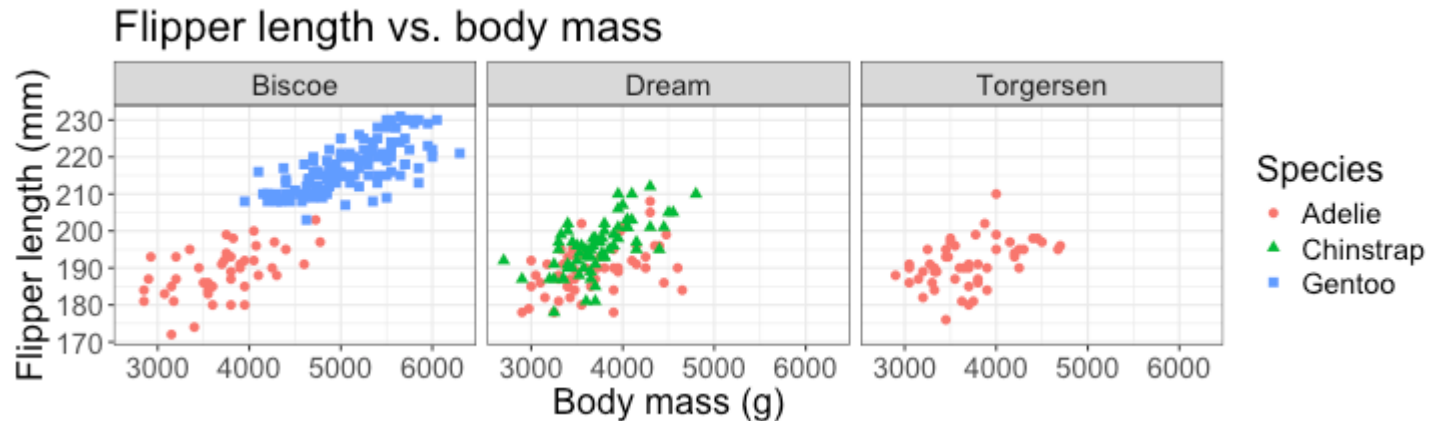
(black + white)

theme_light

theme_minimal

etc.

Making the plot look nice: labels and theme



```
penguins |>
  ggplot(aes(x = body_mass_g,
             y = flipper_length_mm,
             color = species,
             shape = species)) +
  geom_point() +
  facet_wrap(~island) +
  labs(x = "Body mass (g)",
       y = "Flipper length (mm)",
       color = "Species",
       shape = "Species",
       title = "Flipper length vs. body mass") +
  theme_bw()
```

← aesthetics

← geometric object

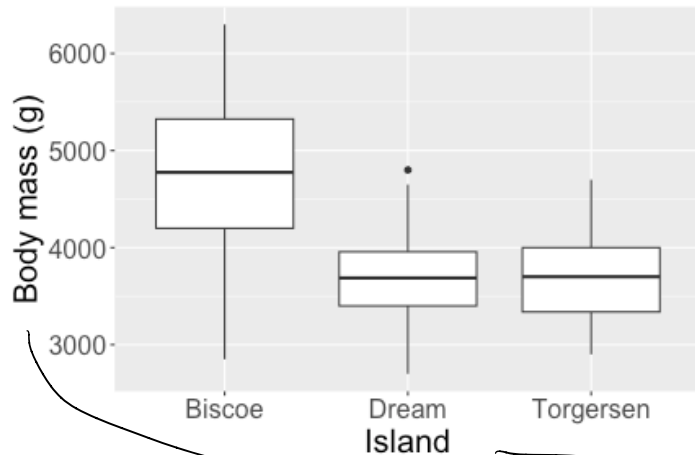
← faceted

← added labels

← change theme

Concept check

Which code created this plot?



```
(A) penguins |>  
      ggplot(aes(x = island,  
                  y = body_mass_g)) +  
      geom_boxplot()
```

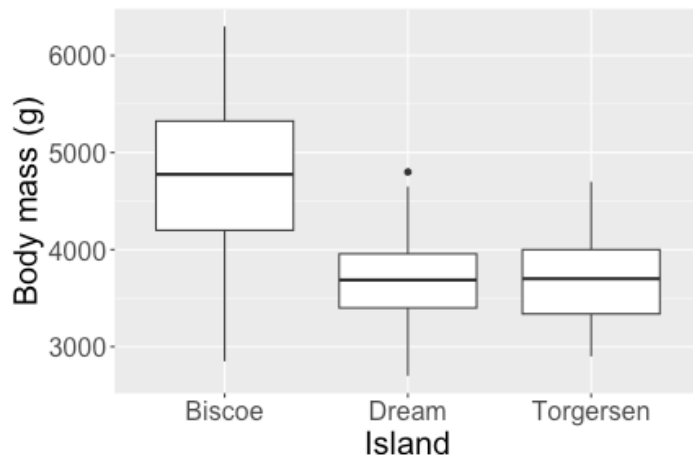
```
(B) penguins |>  
      ggplot(aes(x = island,  
                  y = body_mass_g)) +  
      geom_boxplot() +  
      labs(x = "Island",  
           y = "Body mass (g)")
```

```
(C) penguins |>  
      geom_boxplot()
```

```
(D) penguins |>  
      ggplot(aes(x = island,  
                  y = body_mass_g)) +  
      geom_histogram() +
```

Concept check

Which code created this plot?



Answer: (B)

```
(A) penguins |>  
      ggplot(aes(x = island,  
                  y = body_mass_g)) +  
      geom_boxplot()
```

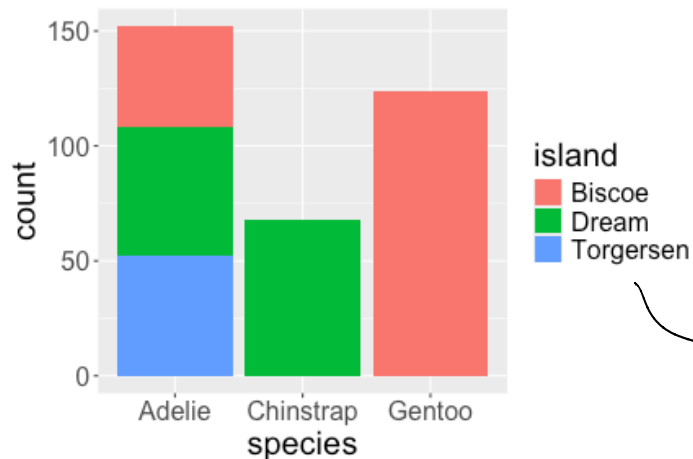
```
(B) penguins |>  
      ggplot(aes(x = island,  
                  y = body_mass_g)) +  
      geom_boxplot() +  
      labs(x = "Island",  
           y = "Body mass (g)")
```

```
(C) penguins |>  
      geom_boxplot()
```

```
(D) penguins |>  
      ggplot(aes(x = island,  
                  y = body_mass_g)) +  
      geom_histogram() +
```

Concept check

Which code created this plot?



(A) `penguins |>
ggplot(aes(x = species))`

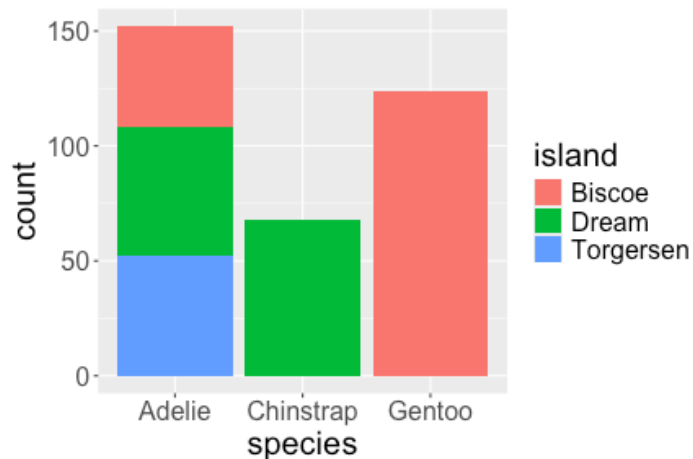
(B) `penguins |>
ggplot(aes(x = species,
fill = island)) +
geom_point()`

(C) `penguins |>
ggplot(aes(x = species,
fill = island)) +
geom_bar()`

(D) `penguins |>
ggplot(aes(x = species)) +
geom_bar()`

Concept check

Which code created this plot?



```
(A) penguins |>  
      ggplot(aes(x = species))
```

```
(B) penguins |>  
      ggplot(aes(x = species,  
                  fill = island)) +  
      geom_point()
```

```
(C) penguins |>  
      ggplot(aes(x = species,  
                  fill = island)) +  
      geom_bar()
```

```
(D) penguins |>  
      ggplot(aes(x = species)) +  
      geom_bar()
```

Answer: (C)

Class Activity

https://sta112-f25.github.io/class_activities/ca_04.html

- + Welcome to work in groups
- + Submit HTML on Canvas when finished
- + Can work on HW 1 if done early